THE MECHANISM OF DIFFUSION IN THE SOLID STATE

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I.—Introduction

The great volume of research now being done on solid-state diffusion, resulting in more than one hundred papers per year, gives some idea of the importance which is attached to understanding the mechanisms by which and the rates at which atoms move with respect to their neighbours in crystalline substances. Diffusion is the dominant step in many familiar metallurgical processes: carburizing, the growth of oxides on metals and precipitates in metals, and sintering, to name only a few. In the limited space available for this review, it is impossible to give a complete bibliography or to discuss all the interesting problems at present under investigation. Fortunately, a bibliography, nearly complete up to the end of 1956, can be quickly compiled from the many recent monographs, symposia, and review articles given as references 1–16. This account refers only to the outstanding papers bearing on each topic discussed. When conflicting data exist, usually only the measurements which seem to the author most likely to be correct are cited.

II.—The Diffusion Coefficient

The diffusion coefficient or diffusivity, D, is defined by Fick's equation:

$$J = -D\frac{dc}{dx}A \qquad . \quad . \quad . \quad . \quad (1)$$

where J is the mass flowing per unit time through a plane of area A, c is the concentration of the diffusing substance, and x is the distance measured normal to the plane. It is assumed here that the concentration is uniform within any plane parallel to the reference plane. Thus, D is the value of the flow rate, J, per unit area when a unit concentration gradient exists at the plane of measurement. D is usually given in units of cm.^{2}/sec.

The measurement of a flow rate under a fixed concentration gradient is difficult in most solid systems. It is more convenient to measure the

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change in concentration with time. When the unidirectional diffusion condition is obeyed, a useful relation can be derived from equation (1) relating concentration with time, t:

$$\frac{dc}{dt} = \frac{d}{dx}\left(D\frac{dc}{dx}\right) \quad . \quad . \quad . \quad (2a)$$

and when D is independent of concentration over the range in the specimen, equation (2a) becomes:

$$\frac{dc}{dt} = D\frac{d^2c}{dx^2} \quad . \quad . \quad . \quad . \quad (2b)$$

These equations can be solved explicitly under certain boundary conditions which are relatively easy to approximate closely in standard experimental procedures. The conditions and solutions are well known, so they are not repeated here (see refs. 1–5). It will be seen that physical phenomena accompanying some diffusion processes make exact application of Fick's equations impossible.

When diffusion is not unidirectional and when the specimen is anisotropic with respect to diffusion, more than one diffusivity is required, since diffusion is a vectorial property. The only cases to be dealt with here relate to cubic crystals, requiring only a single diffusivity, and hexagonal and tetragonal crystals which require two diffusivity values. In the hexagonal and tetragonal crystals the diffusivities are usually taken parallel to the high-symmetry c-axes, and designated D_c, or normal to the c-axes, and designated D_a, since it applies to diffusion parallel to the a-axes.

The diffusivity D_θ, in a direction at an angle θ from the c-axis, in either of these crystal structures is related to D_c and D_a by:

$$D_\theta = D_c \cos^2\theta + D_a \sin^2\theta \quad . \quad . \quad . \quad (3)$$

III.—Diffusion Mechanisms in Metals

Three types of mechanism have been proposed for metallic crystals:

(1) Exchange or rotation mechanisms require a coplanar group of atoms forming a closed ring. If the atoms are numbered successively around the ring in the direction of rotation, the nth member must be a nearest neighbour of the $(n-1)$th and the $(n+1)$th member. When the ring rotates, each atom is replaced by the preceding atom. The simplest ring consists of a pair of nearest neighbours, and the rotation of the pair is called simple exchange (Fig. 1).

(2) In interstitial mechanisms the diffusing atoms move through the interstitial spaces between the atoms occupying the normal lattice sites. The interstitial defect must have a continuity over at least several interatomic distances, although the defect need not centre on the same atom

over this distance. In a special type of interstitial mechanism, designated *interstitialcy* by Seitz,[17] the interstitial defect moves when the interstitial atom squeezes into a nearest lattice site, displacing an atom from that site into another interstitial position. By a repetition of such displacements the defect may move continuously over considerable distances. The crowdion mechanism proposed by Paneth[18] has the properties of an interstitialcy.

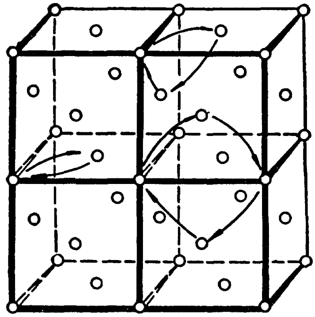

FIG. 1.—Two-, Three-, and Four-Membered Rings in a Face-Centred Cubic Structure. The two-ring mechanism is also called simple exchange.

(3) Vacancy mechanisms require normal lattice sites unoccupied by atoms, which are able to move by receiving atoms from near-neighbouring sites, leaving the site from which the atom jumped vacant. For this mechanism it is possible that vacancies may cluster in groups of two (pairs) or more (triplets, &c.), and that the groups may remain associated through many unit jumps. For a pair to retain its identity, the atom jumping into one vacant site of the pair must originate from a site which is a nearest neighbour of both sites in the vacancy pair.

The diffusion coefficient in equations (1) and (2) measures the relative motion of atoms A and B counter-current to one another in a phase made up of A and B. In a self-diffusion experiment, D measures the relative motion of two isotopes, A^x and A^y, of the same element. Except for variations of atomic volume with changes in composition which may cause small deviations from the assumptions underlying Fick's equations, no major deficiencies are to be expected in applying equations (1) and (2) to metallic systems if diffusion takes places by a rotation mechanism. The rotation of a ring in which all atoms are identical passes undetected. In a ring containing atoms of both types only the net effect of moving A atoms in one direction and B atoms in the reverse direction can be observed. The number and disposition of the lattice sites remains fixed.

The need for a more general scheme or a more sophisticated application of Fick's equations was suggested by Johnson's[19] measurements of the self-diffusion of gold and silver in a silver–gold alloy. He found that the self-diffusion coefficients of silver were substantially greater than those for gold in a 49·2 at.-% gold alloy, and that the average interdiffusion coefficient (D of equations (1) and (2)) over the range 45–55 at.-% gold was about four times greater than the self-diffusion coefficient of gold in the alloy. The fortuitous difference in these values, and the erroneously low self-diffusion coefficients of gold reported by Zagrubsky,[20] permitted Birchenall and Mehl[21] to construct an explanation of Johnson's results in terms of an exchange mechanism in which gold tracer was transported in the alloy as a result of gold/silver atom interchanges, with very little contribution from interchanges between the different isotopes of gold. Although the discovery of the Kirkendall effect, to be discussed below, very soon showed that this type of explanation could not be extended to cover all observations, more recent self-diffusion measurements on both components in α- and β-brasses would have shown the lack of generality of this proposal. Inman, Johnston, Mercer, and Shuttleworth[22] have measured self-diffusion coefficients of zinc and copper in α- and β-brasses. Kuper, Lazarus, Manning, and Tomizuka[23] extended the β-brass measurements to lower temperatures, into the ordered solid solutions. In both α- and β-brasses the self-diffusion coefficient of copper, which is lower than that of zinc, is considerably more than a factor of four below the interdiffusion coefficients given by da Silva and Mehl[24] for α-brass and by Landergren, Birchenall, and Mehl[25] for β-brass. Self-diffusion coefficients of gold (Kurtz, Averbach, and Cohen[26]) and of nickel, and interdiffusion coefficients in gold–nickel alloys (Reynolds, Averbach, and Cohen[27]), over essentially the whole range of compositions from 850° to 925° C., show no constant factor of difference between the lower self-diffusion coefficient and the interdiffusion coefficient. The simple exchange mechanism cannot account for the differences in any of these later experiments.

In solid solutions in which one component is interstitial under equilibrium conditions, it is recognized that the interstitial atom should diffuse through the interstitial space, while lattice atoms diffuse through the substitutional sites. Since the two kinds of atoms do not compete directly for the same sites, two independent diffusion coefficients should be obtained. The thoroughly studied iron–carbon system is an example.

Careful measurements of D for the iron–carbon system have been made by standard chemical methods in austenite. Wells, Batz, and Mehl[28] used a concentration-gradient method requiring the solution of equation (2a), and R. P. Smith[29] used a steady-state method employing equation (1). Similar measurements have been made on ferrite by

Stanley.[30] In these cases the interdiffusion coefficient, D, has been identified as the diffusion coefficient of carbon, D_C, making the justifiable assumption that the mobility of iron is very much less than that of carbon. The diffusivities of iron have been determined in austenite by Mead and Birchenall[31] and in ferrite* by Birchenall and Mehl[32] and by Buffington, Bakalar, and Cohen.[33] All three sets of investigators also measured the self-diffusion of iron in γ-iron of very low carbon content.

Thus, an interstitial diffusion mechanism was clearly recognized for the case of interstitial alloys. The need for more than one diffusion coefficient was acknowledged for this particular type of binary alloy. It is pointed out below that a similar state of affairs existed with regard to diffusion in the ionic crystals, of which halides, oxides, and sulphides had been investigated. Perhaps it is surprising that the need for partial diffusivities for the components of binary substitutional alloys was generally recognized only about ten years ago, as a result of the experiments by Smigelskas and Kirkendall[34] and the theory proposed to explain their observations by Darken.[35]

The Kirkendall Effect

Smigelskas and Kirkendall[34] laid fine molybdenum wires on the plane and parallel sides of an α-brass bar of rectangular cross-section. They electroplated a thick layer of copper over the whole bar so that the wires marked the boundary between the brass and copper. At various stages of the diffusion anneal the couple was removed from the furnace, and a section was cropped off. The distance between the wire markers was measured and found to decrease progressively. The decrease in distance was proportional to the square root of the diffusion time, as shown in Fig. 2. It was observed that a substantial amount of disconnected porosity developed on the brass side of the diffusion interface. The marker displacements were much too great to be attributed to changes in specific volume with changing concentration. Such displacements and the development of porosity are inconsistent with a rotational mechanism. Many suggested sources of experimental error were tested exhaustively by da Silva and Mehl,[24] and the essential conclusions of Smigelskas and Kirkendall were fully confirmed and extended to other binary alloy systems. Although marker displacements and porosity formation do not distinguish uniquely between interstitial and vacancy mechanisms, Smigelskas and Kirkendall

* It is assumed that the measurements on α-iron of good purity would be little affected by the small amount of carbon required to saturate ferrite. The solid solution of carbon in α-iron (ferrite) should not be confused with the spinel oxides derived from magnetite (ferrites) mentioned below.

proposed that vacancies were concentrated on the brass side of the markers because zinc diffused outward faster than copper diffused inward. The markers moved when a macroscopic flow occurred which partially eliminated the excess vacancies. The rest of the excess appeared as porosity. With some elaboration this is essentially the mechanism for marker movement currently accepted. Formation of Kirkendall porosity appears to explain some of the anomalous expansion effects observed in the early stages of sintering mixed powders. Yue and Guy[36] have achieved steady-state diffusion of zinc through an α-brass membrane. Marker movements were observed, but little porosity developed.

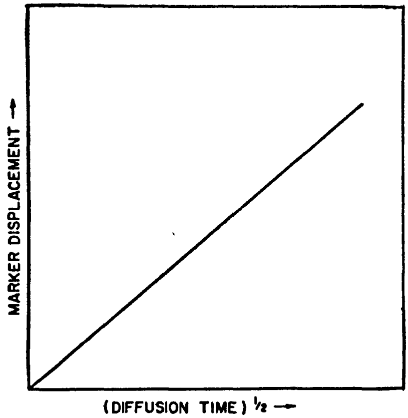

FIG. 2.—Marker Displacement during Isothermal Interdiffusion Plotted against the Square Root of the Diffusion Time (schematic).

Adapting a phenomenological point of view developed by Onsager and Fuoss[37] for solution of electrolytes, Darken[35] derived equations relating the marker displacement, X_m, in time, t, to the partial diffusion coefficients, D_a and D_b, and to the concentration gradient at the marker location:

$$D_a - D_b = \frac{X_m}{2t(\partial N_a/\partial x)} \quad . \quad . \quad . \quad (4)$$

where N_a is the atom fraction of component A and X is the distance measured in the diffusion direction. The partial diffusivities are related to the interdiffusion coefficient by:

$$D = N_b D_a + N_a D_b \quad . \quad . \quad . \quad (5)$$

Darken extended his treatment to a discussion of Johnson's[19] experiments on silver–gold alloys by deriving the relation between the partial diffusivity, the atomic mobility, B, and the thermodynamic activity coefficient, γ, in a form obtained earlier by Dehlinger[38] and others:

$$D_i = kTB_i \left(1 + N_i \frac{\partial \ln \gamma_i}{\partial N_i}\right) \quad . \quad . \quad . \quad (6)$$

where $i = A, B, \ldots$. In an ideal solution, for which $\partial \ln \gamma / \partial N = 0$, by definition:

$$D_i = kTB_i \quad . \quad . \quad . \quad . \quad (7)$$

the well-known Nernst–Einstein equation. Combining equation (6) with equation (5), Darken found:

$$D = (N_b D_a^* + N_a D_b^*) \left(1 + N_a \frac{\partial \ln \gamma_a}{\partial N_a}\right) \quad . \quad . \quad (8)$$

where D^* signifies a self-diffusivity measured in an alloy in the absence of a chemical concentration gradient. He showed that the results of Johnson were internally consistent with this equation when the activity coefficients measured by Wagner and Engelhardt[39] were inserted. Since that time it has been found that the diffusion and thermodynamic data are consistent, within experimental error, with equation (8) for α-brass (Horne and Mehl[40]), β-brass (Landergren, Birchenall, and Mehl[25]), for nickel–gold (Reynolds, Averbach, and Cohen[27]), and for silver–gold (Mead and Birchenall[41]). There is some evidence of real deviations when large amounts of porosity are produced. However, the uncertainty in measuring the interdiffusion coefficient also rises rapidly with production of porosity, so that the deviations are difficult to ascertain. Bardeen[42] called attention to the fact that Darken's essentially thermodynamic treatment contains the inherent assumption that if defects like vacancies or interstitials are necessary for the diffusion process to occur, they are in thermal equilibrium concentration at every point in the diffusing specimen. The reproducibility of diffusivity measurements on materials of good purity, but not exceptional purity, of different origins and prior treatments, suggests that the defects do not deviate extremely from the equilibrium values. On the other hand, the growth of porosity indicates that the equilibrium concentration is exceeded in the penetration zone.

In order for diffusion to proceed in the absence of a rotational mechanism, vacancies or interstitials must be produced in the specimen where needed and eliminated where diffusion tends to concentrate them. The possibility that the external surfaces were the principal sources of vacancies in Smigelskas and Kirkendall's experiment, as first suggested by Seitz,[43] was denied by da Silva and Mehl's[24] observation that the diffusion coefficients and marker displacements did not depend on which of the components, e.g. copper or brass, was on the outside of a diffusion couple with the other material inside.

It became necessary to find other sources and sinks for points defects.[44] It is believed currently that major sources and sinks are at grain boundaries and at jogs in dislocation lines. As jogs move along a dislocation, atom sites can be added to or subtracted from the crystal without a change in the number of atoms available for filling the sites. Looking at the other side of this picture, the diffusion of point defects to dislocation lines is the favoured mechanism for the polygonization process in the recovery of cold-worked metals and for steady-state creep. Unless something of this sort happens, the defects necessary to maintain diffusion when the partial diffusivities differ might be expected to appear as porosity without marker movement.

The experiments of Balluffi[45] and of Resnick and Seigle[46] have shown quite satisfactorily that the porosity is heterogeneously nucleated on inclusions, often oxides, in the penetration zone of a diffusion couple. Supersaturation must be required for this nucleation, so that the dislocation mechanism for removing vacancies cannot be perfectly efficient (Seitz[47]).

Another indication of the importance of moving dislocations as sources of point defects is contained in the work of Buffington and Cohen[48] on self-diffusion in α-iron during straining. Although prior straining had no more than a transitory effect on the diffusivity, straining by compression during the diffusion anneal raised D, roughly in proportion to the strain rate, by a factor as great as 15. Although no satisfactory theoretical explanation has been given of the magnitude of the effect, new experiments on self-diffusion in silver deforming under torsion by Lee and Maddin[48(a)] show even greater acceleration of diffusion during straining. A large reduction in activation energy accompanies increased strain rate.

Generally, markers are inserted only in the initial welded interface. However, Seith, Heumann, and their co-workers (see ref. (4)) have added markers in planes parallel to the initial diffusion interface and have observed the relative displacements. The most complete study is that of Heumann and Walther[49] on silver/gold couples. The difference in mobilities of the two metallic components leads to a non-uniform distribution of vacancies in the diffusion zone. A maximum vacancy concentration occurs on the side of the more mobile component, a minimum on the side of the less mobile component. Between these extremes of vacancy concentration all markers are displaced in the same direction, while beyond, at either end of the diffusion zone, the displacements are in the opposite direction. The specimen surface becomes indented and porosity forms near the plane of the vacancy-concentration maximum, while a bulge in the surface appears in the region of the minimum. The regions of maximum and minimum move farther apart as diffusion proceeds. Markers have their directions of displacement reversed when they are overtaken by one of the extremes of vacancy concentration.

The Temperature-Dependence of Diffusivity

From the preceding discussion it should be evident that purely geometrical arguments do not permit a distinction to be made between an interstitial and a vacancy mechanism, except in the trivial case where interstitial components may be found at equilibrium. The direction in which dislocations move when absorbing defects cannot be observed. Usually a scheme based on vacancies can be restated easily as one based on interstitials. Only one set of experiments has been suggested and performed which gives what may be an unequivocal answer. Before these studies on self-diffusion of sodium are discussed it is necessary to consider the temperature-dependence of the diffusivities.

Within the experimental error, self-diffusion measurements in material kept at constant composition and in thermal equilibrium with its point defects can be described by an Arrhenius-type equation:

$$D = D_0 \exp(-Q/RT) \qquad . \quad . \quad . \quad (9)$$

where T is the absolute temperature, R is the gas constant, and D_0, the frequency factor, and Q, the activation energy, are constants of the material. The single exception reported is for α-iron (Fig. 3), where the data show either a general curvature or a sharp change in slope in the neighbourhood of the Curie point. Birchenall and Mehl[32] drew a single straight line for the whole temperature range, while calling attention to the systematic deviations. Buffington, Bakalar, and Cohen[33] drew their line only through the points above the Curie point. Although the equations of the lines differ substantially, it is evident that the points are from the same population, but are too few and of too little precision to define the behaviour in the Curie-point region. Recent measurements of self-diffusion in α-iron at 50° intervals from 650° to 850° C. by Golikov and Borisov[50(a)] exhibit no anomaly in the Curie-point range. However, the values are roughly three times larger than those just mentioned, paralleling Birchenall and Mehl's average line. Unfortunately, the self-diffusion data of useful precision for cobalt do not extend through the Curie-point range (see Mead and Birchenall[50] for a summary of earlier work on cobalt). This is a problem worthy of careful further investigation.

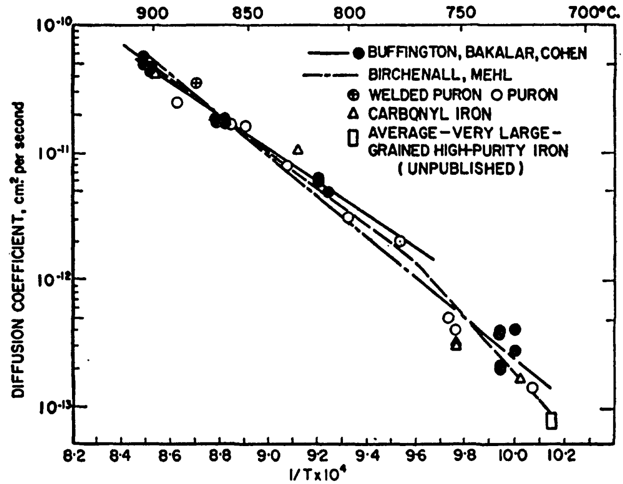

FIG. 3.—Temperature-Dependence of Self-Diffusion in α-Iron. (Includes unpublished results on very large-grained iron by Birchenall.)

The Activation Energy

For a rotation mechanism the activation energy consists entirely of the energy to squeeze the rotating group of atoms past the interfering surroundings, permitting the environment to relax elastically[51(a)] in order to minimize the energy at the saddle point.* For the interstitial and vacancy mechanisms the activation energy consists of the energy to form the defect plus the energy to move it from one equilibrium site to the saddle point, from which it can move to the next equilibrium site. In the jump of the defect, the atom usually must squeeze between some closely packed neighbours. Huntington and Seitz[51] first attempted to calculate the activation energies to be expected for self-diffusion in copper for three mechanisms: simple exchange of nearest neighbours in pairs, interstitial mechanism, and vacancy mechanism. They concluded that the first two required far too much energy, while the vacancy mechanism should be just about right. Zener[52] carried out a new set of calculations for a copper-like model, but allowing larger groups to rotate. Three- and four-membered ring groups can be found in face-centred cubic crystals (see Fig. 1), but only four-rings are possible in body-centred cubic. The four-ring should be much easier than simple

* The saddle point is the highest energy configuration along the available path which the system requires the least energy to travel in the course of a single act of diffusion.

exchange in copper, though still not competitive with the vacancy mechanism. Zener proposed that in the more open body-centred cubic structures, particularly in the softer metals like the alkali metals, this mechanism might be important. Paneth[18] proposed a special type of interstitialcy mechanism for the alkali metals, in which the defect was constrained to move along one of the closest-packed directions as the extra atom displaced a neighbour from a lattice site, and this process was repeated again and again. Because of the apparently low resistance to mechanical relaxation of the neighbours of the interstitial along the close-packed line, this mechanism seemed to be favoured energetically over others, but Herring, in a private communication to Fumi,[53] claims that a very sizeable energy contribution* was omitted from the calculation, which would make this an unimportant mechanism. Fumi[53] and Brooks[54] both apply corrections to Huntington and Seitz's original vacancy calculations which bring them into better agreement with experiment. Machlup[55] reports a lower energy of formation for a vacancy than for an interstitial in sodium, by methods similar to those used by Huntington and Seitz.

In spite of the considerable uncertainties in the difficult calculations of this kind and the dangers inherent in extending the conclusions to materials other than copper or sodium, they represent the principal basis for attributing diffusion in the face-centred cubic pure metals and substitutional alloys to the vacancy mechanism rather than to an interstitial mechanism.

Nachtrieb and Handler[56] found empirically that Q for self-diffusion correlates closely with the latent heat of melting, L_m, for many cubic substances, according to:

$$Q = 16{\cdot}5 L_m \quad . \quad . \quad . \quad . \quad (10)$$

including α-white phosphorus in addition to metals. However, the soft or polarizable lead cannot be made to conform, and the recent values for gold and cobalt do not agree at all well.† A similar relation to this one between Q and the melting temperature T_m is somewhat analogous, since Hildebrand[57] had observed that the entropy of melting, ΔS_m, where:

$$\Delta S_m = L_m / T_m \quad . \quad . \quad . \quad (11)$$

was nearly constant for materials which had the same crystal structure. Nachtrieb's rule is rather more general than the Q/T_m correlation, since Hildebrand's rule would predict somewhat different entropies of melting

* Paneth[18] and Huntington and Seitz[51] neglected the increase in electronic energy arising from the electron gas displaced by an interstitial ion. Fumi estimates that their calculated energies for the formation of interstitials are low.

† Self-diffusion data for metals and a few other elements are summarized in the Appendix (p. 271). The references are included there.

for body-centred and face-centred cubic metals.[58] The correlation with melting properties led Nachtrieb and Handler[56] to postulate a relaxed vacancy mechanism for all these cubic solids.

In the self-diffusion measurements on sodium under a range of applied pressures[79] (discussed below), after the effect of pressure on D_0 was corrected, the logarithm of the self-diffusion coefficients for all temperatures and pressures fell on a single straight line when plotted against the reciprocal of the reduced temperature, T_m/T. Thomas and Birchenall[57(a)] had shown that the diffusion coefficients of a variety of alloying elements in very dilute solutions in lead and copper differ from the self-diffusion coefficient of the pure solvent metal by an amount which correlates approximately with the depression or elevation of the solidus lines in the corresponding binary phase diagrams. In the copper solutions higher diffusivities were observed if the solidus temperature decreased from the melting point of copper, while lower diffusivities applied in a few cases in which the solidus temperature increased. Nachtrieb, Petit, and Wehrenberg[56(a)] pointed out that the increases in self-diffusion in silver, observed by Hoffman, Turnbull, and Hart[58(a)], as a result of small amounts of dissolved lead, aluminium, germanium, copper, or thallium, correlate semi-quantitatively with the depressions of the liquidus/solidus pairs. They measured self-diffusion of silver with four concentrations of palladium (up to 21·84 at.-%) and showed a decrease in diffusivity with increasing solidus temperature, T_s, describing the results for all compositions studied quantitatively by a single equation:

$$D_{\mathrm{Ag}} = 0{\cdot}270 \exp\left(-17{\cdot}75T_s/T\right). \qquad (12)$$

It is evident that there must be a strong resemblance between the average structure in a liquid metal and the immediate neighbourhood of a vacancy in a solid metal. It is not clear whether the extent and nature of this resemblance is the same in all metals. Nor is it clear in what respects the relaxed vacancy described by Nachtrieb and Handler differs from the relaxed vacancy described by others in the language of quantum mechanics.

The Frequency Factor

Because of the exponential form of equation (9), a modest error in Q can lead to a large error in D_0. Thus, for many years when good diffusion measurements claimed little better than 25% uncertainty in individual diffusivities, the reported values of D_0 ranged over many orders of magnitude. It was not until Wert and Zener[59] developed the first satisfactory theory of D_0 that some method of distinguishing between good and bad data became apparent (although it is not foolproof).

Treating self-diffusion as a simple stochastic process, Zener finds:

$$D = \frac{\alpha a^2}{\tau} \quad . \quad . \quad . \quad . \quad . \quad (13)$$

where a is the lattice parameter, α is a geometrical factor related to the arrangement of sites into which an atom may jump, and τ is the average time between jumps. τ may be evaluated by means of a rate-theory equation.

Eyring's[60] rate-theory equation would give:

$$\frac{1}{\tau} = \mathscr{H}\, q \frac{kT}{h} \frac{P^*}{P} \quad . \quad . \quad . \quad (14)$$

where q is the number of equivalent diffusion paths, k is Boltzmann's constant, h is Planck's constant, P^* is the partition function for the activated complex or saddle-point configuration omitting one vibrational degree of freedom, †P is the complete partition function for the atom in the lattice-site configuration, and $\mathscr{H}$ is the transmission coefficient which is usually set equal to unity. $\mathscr{H}$ will be taken as unity here, following Zener.‡

Wert and Zener[59] observe that P^* and P do not contain the same number of degrees of freedom, so that it is not possible to set their ratio equal to an exponential containing a free energy as is usually done. They write $P = P_\nu . P_1$, where P_ν is to be taken as the partition function of a linear oscillator of frequency ν vibrating along the direction leading to diffusion. At high temperatures:

$$P_\nu = \frac{kT}{h\nu} \quad . \quad . \quad . \quad . \quad . \quad (15)$$

When this is inserted in equation (14) the ratio $P^*{:}P_1$ can be related to the isothermal work required to take the diffusing atom over the shortest path from the minimum-energy position in a lattice site to the saddle point for diffusion, while the atom vibrates only in a plane normal to this

† The factor kT/h comes from replacing this vibrational degree of freedom by a translation.

‡ This assumption is probably safe for the simple metallic structures, where the diffusing atom moves in a straight path with the maximum distortion of the structure occurring midway between two equilibrium sites. For more complex structures, where the diffusing atom or ion moves through a tight array of neighbours into a normally unoccupied interstitial site, then changes direction to pass through another tight array to reach a new equilibrium position, the details of the collisions may be important. The transmission coefficient may be considerably less than unity.[61]

path. This is one method for obtaining the free energy, ΔF, of activation. Since $\Delta F = \Delta H - T\Delta S$ and substituting for τ in equation (13):

$$D = q\alpha a^2 \nu \exp(\Delta S/R - \Delta H/RT) \qquad (16)$$

ΔH is equal to Q in equation (9), so that:

$$D_0 = q\alpha a^2 \nu \exp(\Delta S/R) \qquad (16a)$$

Making use of Kê's[62] observation that the shear modulus, μ, decreases linearly with increasing temperature and of an assumption that most of the free energy of activation is used to strain the lattice, Zener concluded that:

$$\Delta S = -\lambda \frac{\Delta H}{T_m} \frac{\partial(\mu/\mu_0)}{\partial(T/T_m)} \qquad (17)$$

would always be a positive quantity. μ_0 is the value of μ extrapolated to absolute zero temperature, T_m is the melting point, and λ is a constant. The size of ΔS was estimated from Köster's[63] measurements of the variation of Young's modulus with temperature for many metals, ν was estimated, and the other constants calculated or obtained from experiments. The values of D_0 from equations (16*a*) and (17) were compared with experimentally determined values. It was found that the most reliable diffusion measurements had values of D_0 lying close to or within the range 0·1–10.

For f.c.c. metals, values close to 0·55 have been obtained for λ in equation (17). For b.c.c. metals, λ lies near unity. Zener initially took this as evidence for a mechanism in b.c.c. metals, probably a ring mechanism, different from the vacancy mechanism generally accepted for f.c.c. metals. By somewhat more elaborate arguments of the same nature, LeClaire[64] came to a similar conclusion. When Shewmon and Bechtold[65] found Kirkendall effects in b.c.c. titanium/molybdenum couples,* Zener[66] suggested that the earlier theoretical developments had omitted a source of entropy difference in the formation of a vacancy in b.c.c. as compared with f.c.c. crystals. The nearest neighbours of a vacancy in f.c.c. metals are in part nearest neighbours to each other. When they attempt to relax to use the volume left by the vacancy, they interfere; the relaxation should therefore be relatively small. In b.c.c. metals the neighbours of the vacancy are not nearest neighbours to each other, so that greater relaxation should occur. The

* Marker movements have been observed in other b.c.c. binary metallic systems. Landergren, Birchenall, and Mehl[25] and Resnick and Balluffi[67] found large displacements and Kirkendall porosity in β-brass. Inman *et al.*,[22] by means of self-diffusion measurements, obtained substantially the same partial diffusion coefficients with $D_{Zn} > D_{Cu}$, just as in α-brass. More recently Yang, Simnad, and Mehl[68] have found marker movements at $-6°$ and $-20°$ C. in potassium/rubidium couples, which should remove any lingering doubts that a ring mechanism can account for diffusion in the alkali metals.

entropy change resulting from the change in vibration frequency from that in a perfect lattice†, ν, to that for an atom next to a vacancy, ν_1, is:

$$\Delta S = ZR \ln (\nu/\nu_1) \qquad . \quad . \quad . \quad (18)$$

where Z is the co-ordination number. With the decrease in strength of binding ν_1 becomes less than ν, so ΔS should be greater for b.c.c. than for f.c.c. metals. Buffington and Cohen[69] had concluded earlier that the greater value of λ in b.c.c. than in f.c.c crystals was consistent with a vacancy mechanism.

Kuper *et al.*[23] proposed an interstitialcy mechanism to explain the observations that in ordered β-brass the mobility of zinc is about twice that of copper, but that antimony has a mobility indistinguishable from that of zinc, even though it is slightly more mobile than zinc in the disordered range. When the structure is ordered, antimony is believed to exchange positions only with zinc. Slifkin and Tomizuka[70] were of the opinion that enough disorder remained at the lowest temperatures to permit a nearest-neighbour vacancy mechanism to operate.

The unusual values for D_0 resulting from Fensham's[71] apparently careful measurements of self-diffusion in tin have given rise to a great deal of speculation. A negative entropy of activation is clearly implied by these results, if they are accepted at face value. Dienes[72] has suggested that the alteration in vibration frequencies of the atoms around the saddle point should make negative contributions to the entropy of activation, and might be sufficiently large to overcome the positive contributions from other sources for a vacancy mechanism. Huntington, Shirn, and Wajda[73] concluded that the activation entropy might be negative for interstitial defects in a copper-like model, but felt that for vacancies the entropy would be positive. Breen and Weertman[74] and Frenkel, Sherby, and Dorn[75] noted the failure of the self-diffusion activation energy to agree with the activation energy for creep, as it does for many metals. Breen and Weertman favoured a ring mechanism for self-diffusion, which would be unable to contribute to creep. Frenkel, Sherby, and Dorn, as well as Nicholas,[76] concluded that the self-diffusion values for tin are not correct. Recently measurements of the diffusion of In^{114} in very dilute solution in tin single crystals have been made by Sawatzky.[77] Indium is rather similar to tin, and its diffusion rate in tin shows none of the strong variation of anisotropy with temperature reported for self-diffusion of tin. D_a is roughly double D_c over the whole range of measurement for In^{114}. The need

† ν is usually estimated from the Debye characteristic temperature of the specific heat. Although this quantity depends on the vibrations in normal lattice sites, the vibration for a vacancy mechanism must be the frequency of vibration of an atom toward a neighbouring vacant site. This approximation is acceptable mainly because of the large approximations already contained in Debye's theory.

is obvious for an experimental check of the results for self-diffusion of tin before further theoretical effort is expended in attempting a rationalization.

If Z is the co-ordination number for a crystal and β is the fractional concentration of vacancies, the probability that an atom has a neighbouring site vacant is $Z\beta$*. If r is the rate at which atoms jump into neighbouring vacant sites, the jump frequency $1/r$ is given by:

$$1/r = Z\beta r \quad . \quad . \quad . \quad (19)$$

If ΔF_f is the free energy of formation of a vacancy:

$$\beta = \exp(-\Delta F_f/RT) \quad . \quad . \quad . \quad (20)$$

and if ΔF_m is the energy to move a vacancy to the saddle-point configuration:

$$r = \nu \exp(-\Delta F_m/RT) \quad . \quad . \quad . \quad (21)$$

ν is defined for equation (15). Substituting the three equations above in equation (13) yields:

$$D = \alpha a^2 \nu \exp[(\Delta S_f + \Delta S_m)/R] \exp[-(\Delta H_f + \Delta H_m)/RT] \quad . \quad (22)$$

For the vacancy mechanism α has a value of unity.

Similar procedures can be followed for the other mechanisms, yielding equations of the same form as equation (22). However, α has different values, and ΔH_f and ΔS_f are zero for ring mechanisms and for interstitial alloys, in which the concentration of interstitial component is determined entirely by the alloy concentration.

Nachtrieb, Catalano, and Weil[78] made precise measurements of self-diffusion of sodium over a range of temperatures. They calculated ΔS from the temperature coefficient of the elastic modulus, on the assumption that the free energy of activation in self-diffusion goes into elastic distortion of the lattice, and also from equation (22), inserting values for α appropriate to each conceivable mechanism successively. Agreement was found only for the vacancy and direct-exchange mechanisms. The latter was ruled out on the energetic grounds discussed earlier. The interstitial, crowdion, and four-ring mechanisms did not give agreement. Nachtrieb, Weil, Catalano, and Lawson[79] extended the self-diffusion measurements on sodium to high pressures. From the variation of ΔF with pressure at constant temperature, they obtained an activation volume, ΔV, given by:

$$\Delta V = \left(\frac{\partial \Delta F}{\partial P}\right)_T = -RT\left(\frac{\partial \ln D/\alpha a^2 \nu}{\partial P}\right)_T \qquad (23)$$

For vacancy or interstitial mechanisms, ΔV must be the sum of a volume change accompanying formation of a gramme-atom of defects,

* This section follows the argument advanced by Zener.[16]

ΔV_f, and a volume change accompanying diffusion, ΔV_m. In either case ΔV_m should be positive. For interstitials ΔV_f must be negative, and for vacancies ΔV_f must be positive. In ionic crystals $\Delta V_f > \Delta V_m$, and if this is the case for sodium the sign of ΔV should distinguish uniquely between vacancy and interstitial mechanisms. Experimentally, ΔV is positive and about half the gramme atomic volume of sodium. Both the sign and the large magnitude strongly favour the vacancy mechanism.

These two studies constitute the only published experiments which offer a choice between vacancy and interstitial mechanisms in metals independent of the energetic calculations outlined above. The conclusions from both procedures are consistent.

In summary, it may fairly be said that there is at present no experimental evidence for a ring or rotation mechanism in any metallic system. Evidence for interstitial diffusion mechanisms is confined almost entirely to systems in which one component occupies interstitial positions as equilibrium sites, i.e. in interstitial solid solutions. In other words, on the evidence now available all diffusion in substitutional solid solutions* may be described by a vacancy mechanism. The recent developments in theory seem to weaken substantially the claims of other mechanisms.

IV.—Diffusion Mechanisms in Ionic Crystals

Because of the large repulsion energy associated with the exchange of two ions of opposite charge, the ring mechanisms have not been seriously considered for diffusion in ionic crystals. The requirements that a local volume of the crystal maintain an overall electrical neutrality makes the description of the interstitial and vacancy defect appear more complex than in the metallic case. The description is complicated somewhat by the more or less parallel nomenclatures deriving from the physical and chemical points of view, and by the several cumbersome sets of symbols devised for writing equations. In the following discussion any symbol not relating to a component of the crystalline phase is identified by (g) to indicate that it belongs to the gas phase.

According to the band theory of solids, the atomic energy levels interact when the ions are brought together to form a crystal. The closely spaced levels fall into discrete energy bands with energy gaps between the bands in many crystals. Insulators must have such gaps, with electrons just filling the band corresponding to the valence electron levels completely, leaving the next higher band, the conduction band, completely empty. The width of the energy gap between the valence and conduction band determines whether observable electronic

* Excluding solid solutions damaged by external radiation.

conductivity can be obtained by heating the crystal below its melting point. The threshold frequency for photoconduction increases as the width of the energy gap increases. When light of frequency higher than the threshold value is absorbed by the crystal, electrons are raised from the valence band into the conduction band. The electrons in the conduction band act as negative carriers of electricity and are often designated free electrons. The empty states in the top of the valence band act as positive carriers of electricity, and are called electron holes. The conduction electrons are written e^-, and the electron holes $\oplus$. Whenever they are created in pairs the result is intrinsic conductivity.

Conductivity may arise in an insulator when an impurity atom replaces a lattice atom or occupies an interstitial position and increases or decreases the electron density. The controlled production of impurity conductors by alloying germanium and silicon with elements of valency not equal to four is a familiar process. Impurity semi-conductors with excess electrons are *n*-type, while those with electron holes are *p*-type.

Although the electronic defects may appear to be incidental, since they do not move ions, they are important in some mechanisms of ionic transport. Certain crystals, however, exhibit purely ionic conductivity. This implies that there is disorder in the ion sites without conduction electrons or holes being available. In 1926, Frenkel[80] proposed that interstitial defects might form leaving vacant sites behind, as shown schematically for AgCl and $PbCl_2$ in Figs. 4 (*a*) and (*b*). As long as the concentrations of the two defects are the same and the interstitial retains the same charge that it had in the lattice site, no free electrons or holes are necessary to keep the region neutral. The interstitials or vacancies, or both, may contribute to ionic conduction.

In 1930 Wagner and Schottky[81] suggested that where the ions are too nearly alike in size for interstitials to form readily, vacancies may appear in cation and anion sites, as illustrated in Fig. 4 (*c*), in such a way that there is an equal deficiency of positive and negative charges per unit volume. This is known as Schottky disorder.

Simple Frenkel and Schottky defects appear only in materials with such a wide energy gap that the formation of electronic defects requires a large energy expenditure. Electrolysis experiments on such materials obey Faraday's laws.

When the crystals are primarily electronic conductors, the defects are associated with deviations from stoichiometric composition. The oxides and sulphides have proved of particular interest in this connection. Wagner's many contributions have been noteworthy in developing an understanding of their transport phenomena. Wagner[82] directs special attention to the pressure-dependence of the composition and of the electrical conductivity.

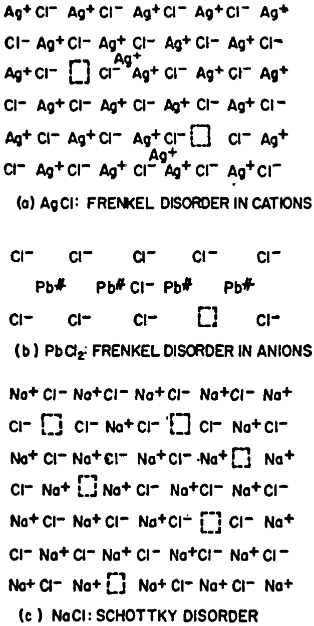

FIG. 4.—Stoichiometric Defects in Ionic Crystals.

For reasons examined below it is believed that zinc oxide contains interstitial zinc ions with their positive charges balanced by conduction electrons. von Baumbach and Wagner[83] show the development of these defects from stoichiometric ZnO by means of the equation:

$$Zn^{++} + O^{=} = Zn_i^{++} + 2e^- + \tfrac{1}{2}O_2(g) \qquad . \quad (24)$$

where the subscript i indicates an interstitial ion formed from a lattice ion, Zn^{++}. Assuming that the defects are in very dilute solution and do not interact appreciably, the law of mass action may be applied in terms of the concentrations. The equilibrium constant for equation (24) is:

$$K^1 = \frac{[Zn_i^{++}][e^-]^2 P_{O_2}^{\frac{1}{2}}}{[Zn^{++}]\,[O^{=}]} \qquad . \quad . \quad . \quad (25)$$

Within the range of composition possible for the ZnO phase, $[Zn^{++}]$ and $[O^{=}]$ do not change appreciably and may be combined with K^1 to give:

$$K = [Zn_i^{++}][e^-]^2 P_{O_2}^{\frac{1}{2}} \qquad . \quad . \quad . \quad (26)$$

Electrical neutrality requires that $[Zn_i^{++}] = \tfrac{1}{2}[e^-]$. The change in the concentration of interstitial zinc ions with oxygen pressure should be given by:

$$[Zn_i^{++}] = \left(\frac{K}{4}\right)^{\frac{1}{3}} P_{O_2}^{-\frac{1}{6}} \quad . \quad . \quad . \quad (27)$$

Similarly the conductivity, $\mathscr{H}$, should be proportional to the concentration of free electrons, which have a much higher mobility than the zinc interstitials in an applied electrical field. Then:

$$\mathscr{H} = (2K)^{\frac{1}{3}} P_{O_2}^{-\frac{1}{6}} \quad . \quad . \quad . \quad (28)$$

Mollwo and Stöckmann[84] suggested that the interstitial zinc ions were only singly ionized according to:

$$Zn^{++} + O^{=} = Zn_i^{+} + e^{-} + \tfrac{1}{2}O_2(g) \quad . \quad . \quad (29)$$

By the same reasoning as above:

$$\mathscr{H} = \text{constant } P_{O_2}^{-\frac{1}{4}} \quad . \quad . \quad . \quad (30)$$

Values of n in $P_{O_2}^{-1/n}$ for the pressure-dependence of conductivity have been observed ranging from 3·6 to 6·0, the most recent and probably the most accurate being 6·0.[85]

Note that the conductivity measurements alone do not distinguish the divalent zinc interstitial from an oxide-ion vacancy, $\square_{O^{=}}$, since the reaction:

$$O^{=} = \square O^{=} + 2e^{-} + \tfrac{1}{2}O_2(g) \quad . \quad . \quad (31)$$

also leads to a pressure-dependence of conductivity:

$$\mathscr{H} = \text{constant } P_{O_2}^{-\frac{1}{6}} \quad . \quad . \quad . \quad (32)$$

A distinction between these mechanisms depends on a demonstration that zinc or oxygen has the greater mobility, or that structural changes occur which are characteristic either of interstitial cations or of vacant anions. Self-diffusion coefficients have been reported for zinc ions but not for oxide ions, so that the direct comparison cannot be made. In 1938 Faivre and Michel[86] found that the lattice parameter of CdO, which behaves electrically like ZnO, increases with increasing cadmium content. This evidence is at least consistent with interstitial zinc ions.

Secco and Moore[87] find that the self-diffusion coefficient of zinc varies as $P^{0.65}_{Zn}$ for 0·2–2·9 atm. pressure of zinc, and that the activation energy is 73 k.cal./mole in 1 atm. pressure of zinc vapour. This pressure-dependence favours a singly ionized zinc interstitial. Roberts and Wheeler[88] quote an activation energy of 89 k.cal./mole for diffusion of zinc in zinc oxide under 1 atm. pressure of oxygen, based on unpublished work by Spicar. It may be that the state of ionization of the interstitial zinc is concentration-dependent, tending to avoid too high a concentration of free electrons when the zinc interstitial concentration increases by lowering the state of ionization.

The case of ferrous oxide (wüstite) is interesting because of the variety and internal consistency of evidence about its structure and diffusion behaviour. In 1929 Pfeil[89] carried out a Kirkendall-effect experiment when he painted NiO or Cr_2O_3 markers on an iron surface and then grew iron oxides on the surface. He observed that the markers remained essentially at the metal/oxide interface, indicating that iron diffused outward through the innermost oxide, wüstite, while oxygen did not diffuse inward at a comparable rate. Later Jette and Foote[90] showed by a careful comparison of X-ray and pycnometric densities that the deviations from stoichiometry in this oxide must be iron-ion vacancies rather than interstitial oxide ions. The vacancy concentrations are always fairly large within the stable range of this phase, so that dilute-solution approximations that work for more nearly stoichiometric oxides need not apply to wüstite. Himmel, Mehl, and Birchenall[91] measured self-diffusion coefficients of iron ions in wüstite over a range of temperatures and compositions. Using a theory of oxide growth developed by Wagner,[82] they showed that growth-rate constants in agreement with those measured by Davies, Simnad, and Birchenall[92] could be calculated, assuming that iron-ion diffusion is much faster than oxide-ion diffusion.

The defects in wüstite may be described by either:

$$2Fe^{++} + \tfrac{1}{2}O_2(g) = 2Fe^{+++} + O^{=} + \square_{Fe^{++}} \quad . \quad (33a)$$

or

$$\tfrac{1}{2}O_2(g) = 2\oplus + \square_{Fe^{++}} + O^{=} \quad . \quad . \quad (33b)$$

According to equation (33*a*), increasing the oxygen:iron ratio in wüstite produces cation vacancies, $\square_{Fe^{++}}$, with the neutrality maintained by the oxidation of two ferrous ions to the ferric state for each vacancy formed. Alternatively, the positive charges may be described as electron holes, as in equation (33*b*). The former, or chemical, description and the latter, or physical, description are equivalent in the sense that the trivalent iron state may migrate from one ion to another fairly easily. But it is not a completely free charge, for Kuper[93] has shown that an activation energy is required for its motion, i.e. the electron hole may be bound to the cation vacancy. In the chemical sense, intrinsic conductivity might be compared with a disproportionation reaction in which ions in an intermediate valence state go into a mixture of higher and lower valence states. The ease with which the electronic defects can be formed depends upon the relative stabilities of the various valence states. In the physical manner of speaking, promoting an ion from a divalent state to a trivalent state of low stability involves exciting an electron through a large energy gap. More emphasis has been placed on this dual viewpoint than might seem necessary, because it suggests a way to rationalize the data reported on self-diffusion of cations in oxides which have the NaCl structure. The scheme to be proposed here suggests, in turn, the kind of measurements likely to be most useful.

Cation Diffusion in Oxides with the NaCl *Structure*

Fig. 5 summarizes the self-diffusion coefficients of the cations in FeO, CoO, NiO, MgO, and CaO. The results for MgO, CaO, and NiO were recorded at constant oxygen pressure, while those for CoO and FeO are reported for constant composition. When self-diffusion measurements are carried out at constant composition, the activation energies should reflect only the energies necessary to move the ionic or electronic defects, whichever are the more difficult. When constant pressure of one of the components is employed, electronic and ionic defect concentrations should vary with the temperature. Therefore, the activation energy should include the enthalpy of formation as well as of motion.

The enthalpy of formation of defects may be measured alone by direct determination of the deviations from stoichiometric composition over a

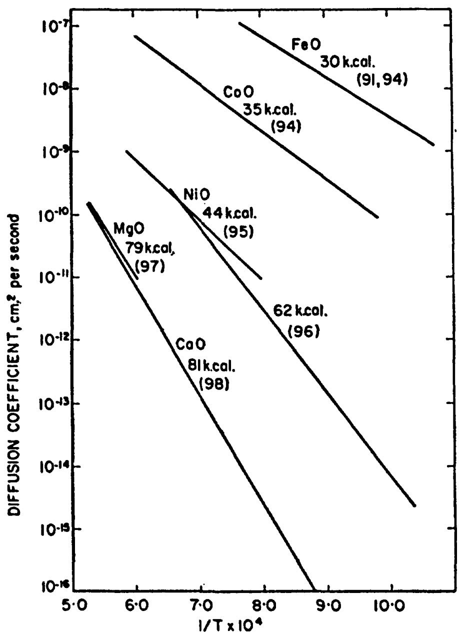

FIG. 5.—Self-Diffusion of Cations in Oxides having the NaCl Structure. Numbers in parentheses are literature references.

range of temperatures at constant partial pressure of the components in the gas phase. Unless the deviations are large, the measurements are difficult; consequently they are available only for a few oxides.

For the reaction:

$$\tfrac{1}{2}O_2(g) = O^{=} + \square_{M^{++}} + 2\oplus \quad . \quad . \quad (34)$$

ΔH_f is the enthalpy to form one mole of cation vacancies plus two moles of electron holes. The enthalpy associated with the formation of one mole of electron holes is $\Delta H_f/2$.

It is probable that too little attention has been given to equilibration of the material in the atmosphere in which diffusion measurements are to be carried out. To a lesser degree this oversight may be true of electrical conductivity measurements. For maximum utility, determinations of both diffusivity and conductivity should be made under conditions of constant partial pressure or of constant composition—preferably both.

Table I makes use of the scheme just proposed. Column 1 lists the activation energy of diffusion measured at constant composition, which should be ΔH_m; column 2 the activation energy of diffusion measured

TABLE I.—*Energies Obtained from Diffusion, Conductivity, and Composition Measurements on Oxides with the* NaCl *Structure**

Oxide	ΔH_m	$\Delta H_m + \Delta H_f$	ΔH_μ	$\Delta H_\mu + \tfrac{1}{2}\Delta H_f$	ΔH_f
FeO	30 (91, 94)	−10 (91)	2 (93)	—	−28 (91) −17 (103)
CoO	35 (94)	—	—	12 (100)	5·1 (94)
NiO	—	44 (95) 62 (96)†	2·3 (99)	18 (83) 21 (101)	—
MgO	—	79 (97)	—	40 (102)	—
CaO	—	81 (98)	—	43 (98) 37 (102)	—

[Literature references are given in parentheses]

*Enthalpies are in k.cal./g.-atom of vacancies or holes.

†Dr. Lindner reports that his latest equation for self-diffusion of Ni^{63} in NiO is $D = 1{\cdot}5 \times 10^{-3}\exp(-61{,}400/RT)$ cm.^{2}/sec. (Private communication, 30/1/58).

under constant partial pressure of oxygen, which should be $\Delta H_m + \Delta H_f$; column 3 the activation energy of electrical conductivity measured at constant composition, which should be the enthalpy of activation for the motion of electron holes, ΔH_μ; column 4 the activation energy of electrical conductivity measured at constant oxygen pressure, which should be $\Delta H_\mu + \tfrac{1}{2}\Delta H_f$. Column 5 lists the values of ΔH_f resulting from measurements of deviations from stoichiometric composition at constant oxygen pressure over a range of temperatures.

In identifying the enthalpies in this way it is assumed: (*a*) that the ionic conductivity does not contribute significantly to the electrical measurements; (*b*) that the conductivities are volume conductivities not dependent on surface states and the diffusivities do not have grain-boundary contributions; and (*c*) that the electron concentration derives from reactions which form cation defects with no significant contribution from intrinsic or impurity conductivity. In FeO and CoO the deviations from stoichiometric composition are sufficiently large for any intrinsic or impurity contributions to be overwhelmed. For MgO and CaO the forbidden-band gap is so great that intrinsic conductivity is probably negligible even at the highest temperatures. Hauffe and Tränckler[104] found that CaO formed oxide vacancies at low oxygen pressures and calcium vacancies at higher pressures. Similar behaviour may apply for MgO. Mansfield[105] concluded that electron holes conduct with an activation energy at constant pressure of 46 k.cal./mole, while Yamaka and Sawamoto[106] favour excess electron carriers with an activation energy of 53 k.cal./mole. Although there is a possible danger of ionic defects contributing through Frenkel or Schottky disorder, Lindner[107] reports that the transference number of calcium in CaO does not exceed 0·01 up to 1300° C.

Moore[108] has attempted a calculation of ΔH_f at 25° C. for FeO, CoO, and NiO. The formation of vacancies not dissociated from the electron holes was estimated to require —56, —23, and —14 k.cal., respectively. The values are in the order expected. The magnitudes differ from those in Table I because the oxygen is referred to standard temperature and pressure rather than to equilibrium pressure. Since small differences between large numbers are involved, the difficulties of such a calculation are considerable.

If these oxides may be approximated by the model of a simple ionic crystal, the following are to be expected: (i) The activation energy for the motion of the cations should not differ greatly from one oxide to another because the cation:anion size ratios are similar. (ii) The enthalpy of formation, insofar as it derives from removal of a divalent cation from the lattice, should show little dependence on the particular oxide since all cations are divalent. (iii) The enthalpy of formation of the electron holes to keep the crystal electrically neutral in the neighbourhood of a cation vacancy should be increasingly positive as the energy gap between valence and conduction states increases (or with the difficulty of making a trivalent ion from the divalent cation).

The data are not sufficiently complete or precise to demonstrate any of these things, but they are clearly consistent with the model. The heat of formation of vacancies and holes is negative in wüstite where iron has a preference for the trivalent state (note that wüstite is unstable below 570° C.). The defects in cobalt oxide have a small positive heat of formation. The trivalent state is observed frequently for cobalt, though apparently less commonly than the divalent state. In nickel oxide, for which a normal trivalent state is not known (although a tetravalent state of low stability has been reported), the enthalpy of formation appears to be about 30 k.cal./g.-atom of vacancies. As might be expected, it requires considerably more energy to make and move the defects in magnesium and calcium oxide. The values for these last two oxides must be considered indistinguishable at this stage of development. The possibility of a high enthalpy to move electrons is not ruled out for these oxides.

The Spinel Oxides

Self-diffusion measurements on cations in spinels are beginning to accumulate in substantial number, particularly on the ferrites which are of some interest for electronic applications. The prototype, magnetite or Fe_3O_4, illustrates one aspect of the problem in these materials. The unit cell consists of 32 oxide ions in face-centred-cubic packing with one eighth of the 64 tetrahedral interstices and one half of the 32 octahedral interstices occupied. The ideal arrangement pictured for magnetite (the inverse spinel arrangement) shows half the ferric ions in the tetrahedral sites, with the other half of the ferric ions and all the ferrous ions occupying the octahedral sites. Three distinguishable kinds of disorder can be imagined: (*a*) randomness in the placement of the 8 divalent and 8 of the 16 trivalent ions on the normally occupied octahedral sites; (*b*) reduction of some of the trivalent ions in tetrahedral sites to the divalent state, while an equivalent number of divalent ions in octahedral sites are oxidized to the trivalent state; (*c*) occupation of some of the normally unoccupied sites. A variant of (*c*) would arise from a deviation from stoichiometric composition changing the ratio of trivalent to divalent iron ions from 2:1 and introducing vacancies or interstitials.

Disorder of the (*a*) and (*b*) types is all that is needed for electrical conductivity. Disorder of the (*c*) type is required for diffusion. Both types are probably present at high temperatures, but not enough detail is known to be of much use in establishing diffusion mechanisms. It is clear that rather tortuous paths must be followed from one site to another crystallographically equivalent to it.

In the case of mixed spinels, valence states may be more strongly associated with particular ions. Electron transfer may be more difficult and may, in turn, make atom movement more difficult. Many of the spinels, particularly of the transition metals, may have stable phase fields permitting a wide variation in the ratio of the two cations. The diffusivities certainly depend on all the variables listed. Fig. 6 summarizes the results currently available. The large differences in self-diffusion coefficients of iron in the ferrites of different composition suggest that control of the growth rates of oxide layers on steels may, in the absence of a stable wüstite phase, be greatly modified by changing the composition of the product spinel phase.[109]

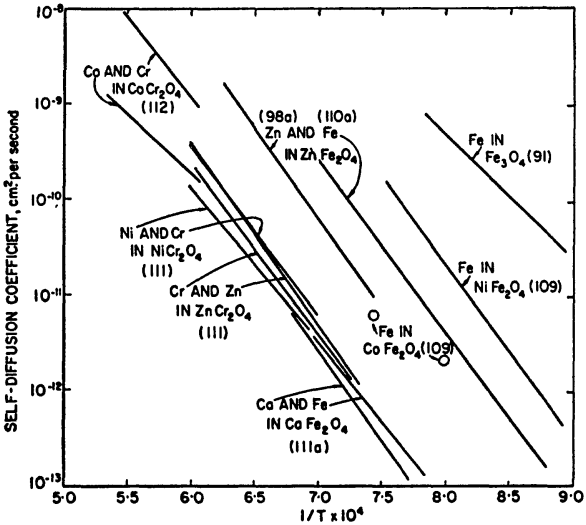

FIG. 6.—Self-Diffusion of Cations in Some Spinel Oxides and in Calcium Ferrite Which is Not a Spinel. Numbers in parentheses are literature references.

Some idea of the likelihood of finding disorder of various kinds in the spinel structures may be obtained by extending the methods developed by Romeijn,[113] and others.[114] Reference should also be made to the recent work of Flood and Hill,[115] in which the thermodynamic properties of some spinel ferrites at high temperatures are found to be consistent with disorder in the cation sites.

V.—DIFFUSION MECHANISMS IN COVALENT CRYSTALS

Self-diffusion measurements have been reported for covalent elements only in a few cases: germanium, carbon (the graphite structure only), α-white phosphorus, and sulphur.[116, 117] Of these crystals germanium alone, which has the diamond cubic structure, is purely covalent. α-white phosphorus[118,119] is a molecular crystal made by packing P_4 molecules in a cubic α-manganese structure by means of van der Waals force. In this case presumably the P_4 molecules diffuse by a mechanism not yet specified.

Dienes[120] calculated activation energies for reasonable models of vacancy, interstitial, and direct-exchange mechanisms of diffusion in graphite, obtaining values of 191, 417, and 90 k.cal./mole, respectively. On this basis, the volume diffusion is considered to be predominantly by exchange. Self-diffusion in graphite was measured in the range 1835°–2370° C.[121] A mixture of volume and grain-boundary transport was observed. If an activation energy of 90 k.cal./mole is assigned to the volume-diffusion mechanism, the grain-boundary diffusion activation energy is 75 k.cal./mole.

Kanter[122] measured self-diffusion coefficients in natural crystals of graphite from 1995° to 2347° C. and found volume diffusion predominant above 2150° C. He reports a diffusivity of $40f^2 \exp [(-163{,}000)/RT,]$ where the geometric factor f has an estimated value between 0·1 and 0·6. The diffusion direction is assumed to be parallel to the hexagonal sheets. Below 2150° C. grain-boundary contributions cannot be neglected. Kanter corrected Dienes's calculations of activation energies for the various mechanisms, using a more acceptable value (170 k.cal./g.-atom) for the heat of sublimation of graphite (instead of 124 k.cal./g.-atom used by Dienes). The vacancy mechanism gives 263, interstitial 467, and exchange 113 k.cal./mole. Kanter points out that the energy for vacancy formation would be reduced if relaxation were included in the model. However, it hardly seems likely to overtake the exchange mechanism if the calculations are essentially correct.

Measurements of self-diffusion in germanium[123] have been made, using a precision grinding technique. An activation energy of 68·5 k.cal./mole was observed. Diffusion of trace impurities in germanium and silicon has been studied by several groups using the grinding method, and also by following the displacement of *p–n* junctions which can be located by electrical probing of one sort or another. Both germanium and silicon crystallize in the diamond cubic structure with the valence bonds of the central atom directed toward four nearest neighbours which lie at the corners of a regular tetrahedron, equidistant from the central atom. The materials can be obtained in a state of extraordinary purity by zone refining, and in a state of unusual crystallographic perfection by careful growth of crystals by slow withdrawal from the melt.

The replacement of a tetravalent germanium or silicon atom by a trivalent element, such as boron, leaves one covalent bond unsaturated.

The electron missing from the valence band conducts as an electron hole. If the solute has a valence lower than three and replaces a solvent atom, a proportionately greater number of holes per solute atom will be made. When a pentavalent atom such as arsenic or antimony displaces germanium or silicon, electrons are left over for the conduction band after the bonds are made. The distribution of regions with positive or negative carriers in excess is very important in the technology of electronic devices. Since many desirable configurations can be achieved in practice by the controlled diffusion of solute elements into germanium or silicon wafers, much attention has been devoted to the study of this process. It constitutes one of the few instances in which diffusion measurements may be applied directly, rather than indirectly through increasing the degree of understanding of more complicated processes.

Tables II and III summarize the diffusion results in silicon and germanium. Because of the low concentrations of solute and the low diffusion rates in many of these studies, disagreement between different investigations of the same system is often relatively large. Nevertheless, trends appear to be present.

In both silicon and germanium, monovalent metals lithium, copper, and gold and the transition element nickel diffuse much more rapidly than solutes from Groups II, III, and V. Copper and lithium have been shown to move as positive ions, suggesting that they and the other mobile atoms take up interstitial positions. Mobilities from the electrical measurements agree with diffusion mobilities when compared

TABLE II.—*Diffusion in Silicon*

Solute	Approx. at 1100°C., D cm.²/sec	D_0, cm.²/sec.	Q,k.cal./mole	Ref.
Li	1×10^{-5}	$1{\cdot}9 \times 10^{-3}$	14·7	124
	2×10^{-5}	$9{\cdot}4 \times 10^{-3}$	18·1	125
		$2{\cdot}3 \times 10^{-3}$	15·2	126
Fe	4×10^{-6}	$6{\cdot}2 \times 10^{-3}$	20·0	126*a*
Au	8×10^{-7}	$1{\cdot}1 \times 10^{-3}$	25·8	126*a*
,,	10^{-7}	—	—	127
Al	2×10^{-12}	8·0	80	128
,,	2×10^{-12}	4·8	77·4	128*a*
B	10^{-12}	—	~ 71	127
Ga	4×10^{-13}	3·6	81	128
B, P	4×10^{-13}	10·5	85	128
Sb	10^{-13}	—	~ 71	127
In, Tl	9×10^{-14}	16·5	90	128
As	4×10^{-14}	0·32	82	128
Sb	2×10^{-14}	5·6	91	128
Bi	$< 10^{-14}$	$1{\cdot}03 \times 10^{3}$	107	128
Si	—	—	~110 (estimated)	128

TABLE III.—*Diffusion in Germanium*

Solute	Approx. at 800°C., D cm.^{2}/sec.	D_0, cm.^{2}/sec	Q,k.cal./mole	Ref.
Ni	5×10^{-9}	0·8	21	129
Cu	3×10^{-9}	—	< 5	130
Li	1×10^{-9}	$2·5 \times 10^{-3}$	11·8	126
,,	1×10^{-9}	$1·3 \times 10^{-3}$	10·7	125
As, Sb	4×10^{-11}	0·71	51	131
As	4×10^{-11}	—	—	132
Sb	3×10^{-11}	1·3	52·1	133(*a*)
,,	2×10^{-11}	10	57	132
,,	2×10^{-11}	1·2	53	133
As	1×10^{-11}	2·1	55	133
P	8×10^{-12}	—	—	132
Zn	6×10^{-12}	—	—	131
In	2×10^{-12}	—	—	131
Zn	1×10^{-12}	—	—	132
B	6×10^{-13}	—	—	132
In	2×10^{-13}	—	—	132
,,	1×10^{-13}	0·15	66	133
Ga	1×10^{-13}	—	—	132
Ge	1×10^{-13}	7·8	68·5	123

by means of the Einstein relation.[126] Recent evidence of a strong dependence of diffusion of copper in germanium on the perfection of the crystal[134] indicates that a simple interstitial mechanism is not a full description. Apparent diffusivities of the order of 10^{-8}cm.^{2}/sec. were observed at 710° C. in the best crystals, compared with 10^{-5}cm.^{2}/sec. in crystals containing low-angle boundaries.

The interstitial positions in the diamond structure can accommodate atoms as large as the lattice atoms without distortion, on a rigid sphere model. Whether a solute atom goes into a lattice or an interstitial site must depend more on its ability or inability to form tetrahedral covalent bonds than on its size, within reasonable limits. For interstitial atoms the open channels parallel to $<110>$ seem to offer paths of easy motion, consistent with the low activation energies reported for the diffusion of nickel, copper, and lithium.

The other elements studied apparently replace germanium or silicon on lattice sites. Such an interpretation is consistent with the observed changes in the population of positive or negative electronic-charge carriers. If the substituted atom has less than four valence electrons, one or more positive holes form in the crystal. If the atom acquires four complete bonds it has a residual negative charge and an effectively increased size. For a substitutional element of valence greater than four, electrons are added to the crystal with a shrinkage of size compared with the atomic radius. Use has been made of the greater size of the negative ions in comparison with the positive ions by Dunlap[132] to explain the fact that antimony and arsenic diffuse faster in germanium than do the Group II and III elements. In silicon, however, arsenic, antimony, and bismuth diffuse more slowly than elements from the lower groups. Fuller and Ditzenberger[128] note that the heavier elements show somewhat higher activation energies than other elements from the same group.

On the basis of thermal-quenching experiments on germanium with safeguards against contamination, Mayburg[135] finds a heat of formation of 46·1 k.cal./mole for the diffusing defects. Subtracting this from the self-diffusion activation energy leaves only 22·4 k.cal./mole as the activation energy for mobility of the quenched-in defects. This value is so low that substantially no defects should have been retained at the cooling rates employed. As a result it was proposed that self-diffusion did not occur by a vacancy mechanism, but by a ring mechanism. Logan,[136] in similar quenching experiments, gave an appreciably lower heat of formation, so that defect mechanisms may not be excluded with complete certainty.

In order to explain the structure sensitivity for diffusion of copper in germanium, van der Maesen and Brenkman[137] proposed an interstitial/substitutional equilibrium. Frank and Turnbull[138] pointed out that such a model requires, in addition, a supply of vacancies to convert interstitial copper atoms to substitutional atoms. Such a supply would be available near dislocations and free surfaces. The model appears to be capable of explaining the observations when reasonable magnitudes are assumed for the parameters.

When the Group IV atoms are replaced by A atoms on one set of alternate sites and B atoms on the other set, so that each A has four nearest neighbours of B and vice versa, the structure is called the zinc-blende or ZnS structure. Numerous pairs of elements from Group III and Group V, Group II and Group VI, and Group I and Group VII, as well as pairs from Group IV (e.g. SiC), crystallize in this structure. Although the binding must be predominantly covalent to maintain such dilute packing, presumably the ionic character increases as the elements are taken from more widely separated groups.

Self-diffusion measurements in InSb, and GaSb by Eisen and Birchenall[137(a)] showed significantly greater diffusivities for indium and gallium than for antimony. On this basis a mechanism involving simple exchange or ring rotations was ruled out. The participation in diffusion of more tracer atoms plated on the surface than the normal deviation from stoichiometric composition would allow, made an interstitialcy mechanism seem unlikely. Following a suggestion of Slifkin and

Tomizuka[138(a)] that in such highly ordered structures a nearest-neighbour vacancy mechanism should give identical diffusivities for both components, a mechanism was adopted based on vacancies restricted to move through antimony sites or through those of the Group III element only. Lidiard[139] has pointed out that a nearest-neighbour vacancy mechanism will permit unlike partial diffusivities. On the basis of the very low values of D measured, it is probable that not many atoms need be in the wrong kind of site at any instant. Perhaps choice between a first- and second-nearest-neighbour vacancy mechanism could be made if the vacancy concentration and the permissible deviation from stoichiometric composition were known, but it does not seem feasible at this time.

Valenta and Ramasastry[139(a)] have shown that self-diffusion coefficients in germanium are raised by an n-type impurity (arsenic) and decreased by a p-type impurity (gallium). Assuming that the solubility of impurity of one type is increased by the presence of impurity of the other type and decreased by impurity of the same type, it is concluded that vacancies act as electron acceptors and are responsible for self-diffusion in germanium.

Diffusion data on a wider variety of intermediate phases in which the structures and binding differ greatly will be required in the years ahead as an effort is made to expand the range of materials that may be prepared in a controlled way and applied for practical ends. It is difficult to prepare many intermediate phases from the melt in a simple manner. Growth of the phases by solid-state diffusion should often be the preferred method.

VI.—Boundary and Surface Diffusion

Surface diffusion was inferred from the observations on crystal growth by Volmer and his co-workers. They found that crystals growing from melts often showed rapid accretion at protruding corners not in direct contact with the source of supply. Crystals of zinc, cadmium, and mercury[140] grew in atomic beams preferentially at some points where the rate of growth greatly exceeded the rate of arrival of vapour atoms. Surface migration was required to deliver atoms to the growing points from the broad receiving surfaces of the crystals.

Because thermionic emission of a tungsten filament is sensitive to the concentration of adsorbed atoms, this property has been used to follow the surface diffusion of adsorbed films of sodium, caesium, barium, and thorium. Surface diffusivities of thorium on tungsten were obtained by Brattain and Becker.[141] On tungsten filaments of varying grain size, Fonda, Young, and Walker[142] measured the rate of diffusion of thorium inward along the grain boundaries, while Dushman and Koller[143] and Clausing[144] measured the accumulation of thorium on the surface by diffusion outward along grain boundaries. Langmuir[145] calculated equations for these processes and volume diffusion as follows:

$$\left.\begin{aligned} &\text{Surface: } D_s = 0{\cdot}47 \exp(-66{,}400/RT) \\ &\text{Grain boundary: } D_b = 0{\cdot}74 \exp(-90{,}000/RT) \\ &\text{Volume or lattice: } D_l = 1{\cdot}0 \exp(-120{,}000/RT) \end{aligned}\right\} \quad . \quad (35)$$

The essential feature, the greater ease of diffusion along grain boundaries and, *a fortiori*, along surfaces, is evident in the equations.

The only example approaching tungsten/thorium in completeness is the self-diffusion of silver. In this case it has been shown clearly that surface and grain-boundary self-diffusion are both anisotropic for at least some orientations of surfaces and boundaries. Average surface self-diffusion measurements were reported first by Nickerson and Parker[146] to be given by:

$$D_s = 0{\cdot}16 \exp(-10{,}300/RT) \quad . \quad . \quad (36)$$

However, their value of about 4×10^{-5} cm.^{2}/sec. at 350° C. is very much greater than the value of $2–6 \times 10^{-9}$ cm.^{2}/sec. reported at 425° C. by Winegard and Chalmers.[147] The latter lies below the extrapolated grain-boundary diffusivities, and is probably much too low.

Surface-diffusion measurements are always open to the usual questions about surface cleanliness. The preparation and maintenance of impurity-free surfaces is necessary to establish true surface diffusivities. Surface-tension studies have shown the sensitivity of silver surfaces to oxygen contamination. Furthermore, studies of mobilities of foreign atoms on surfaces have little significance unless it can be shown that the substrate remains nearly constant in its configuration. One of the more powerful tools, although very limited in the range of materials that can be investigated, is the field-emission microscope.[148] With this instrument surface changes arising from contamination can be observed, as well as those resulting from surface migration. The anisotropy of surface diffusion inferred from such studies has been confirmed by autoradiographic work on silver surfaces by Winegard,[149] who reports the highest diffusivity to be along $\langle 110 \rangle$ in $\{110\}$. Hackerman and Simpson[150] found that self-diffusion on a copper surface was anisotropic on $\{100\}$, where it is faster along $\langle 110 \rangle$ than along $\langle 100 \rangle$, and on $\{110\}$, but not on $\{111\}$. The average diffusivity was different on the three different surface planes. The major effort in surface-diffusion studies still lies ahead.

Preferential grain-boundary penetration into a metal may be observed on a single section taken normal to the initial plane interface, on a series of sections taken parallel to the initial interface, or on a single oblique section making a slight angle with the initial interface. The first and second methods are somewhat more readily adapted to quantitative use. In the first case chemical-etching techniques may be found which yield concentration contour lines. However, autoradiography* may be used in any case if the diffusing element is radioactive.

Smoluchowski and his co-workers carried out a series of grain-boundary diffusion measurements on columnar copper. Because the [100] directions were parallel in neighbouring grains, the boundaries were of the tilt variety, which can be described by a single angle of disorientation between like planes in the adjoining grains. For the diffusion of silver in copper,[151] silver and zinc[152] in copper, and for self-diffusion of iron[153] in silicon iron oriented with parallel [100], no grain-boundary diffusion was observed until the angle between adjacent grains reached 15°–20°; then grain-boundary diffusivity increased with increasing disorientation. From 15° to 35° the diffusivity was found to vary with direction within the boundary.

Hoffman and Turnbull[154] measured grain-boundary self-diffusion in silver, reporting as an average for polycrystalline specimens independent of grain size:

$$D_b = 0{\cdot}03 \exp\left(-20{,}200/RT\right) \text{ cm.}^2/\text{sec.} \tag{37}$$

The grain-boundary, surface, and volume self-diffusivities are given in Fig. 7 for silver. The trend is the same as for thorium on tungsten. Turnbull and Hoffman[155] observed that boundary self-diffusivity of silver depended on the relative orientation of grains (rotated about [100] by 9°–28°) and on the orientation of the boundary itself in a 9° boundary. When Burgers's[156] dislocation model of a tilt boundary was applied it was found that for $\theta < 28°$ the results could be described by a diffusivity for motion along a dislocation pipe, D_p, which is independent of dislocation density:

$$D_p = 0{\cdot}14 \exp\left(-19{,}700/RT\right) \text{ cm.}^2/\text{sec.} \tag{38}$$

An ingenious experiment by Hendrickson and Machlin[157] measured the diffusivity through isolated dislocation pipes. Dislocations were formed in predictable density in a sheet of silver by bending through a known angle about an axis normal to the sheet. The density of dislocations running normal to the sheet was confirmed by etching. Active silver diffused through the pipes from the plated front face and

* Careful calibration of films must be made when they are to be used as quantitative detectors of nuclear radiations. The useful exposure range is often not in the region where the light transmission of the exposed emulsions decreases in a simple linear or exponential way with total intensity of the nuclear radiation absorbed in the emulsion.

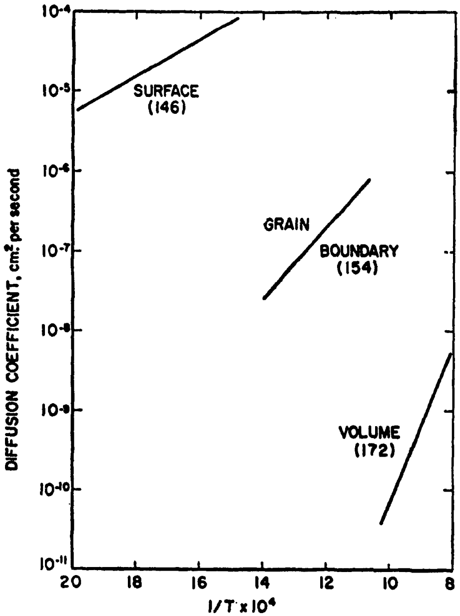

FIG. 7.—Volume, Grain-Boundary, and Surface Self-Diffusion in Silver. Numbers in parentheses are literature references.

spread out on the back, where it was counted. The diffusivity per pipe at 450° C. agreed very well with the results of Turnbull and Hoffman. At 450° C. D_p is greater than D_l by a factor between 10^6 and 10^7.

There seems little doubt that more rapid boundary diffusion extends right down to the smallest angles. However, observation becomes more difficult at low angles because the dislocation density decreases and the total amount of material transported through the boundaries decreases proportionately.

A theory developed by Fisher[158] and extended by Whipple[159] makes it possible to separate approximately the volume and grain-boundary contributions when both occur together. Fisher assumed a model of thick, slowly conducting slabs separated by thin slabs of uniform width δ, with material entering from a plane surface cut normal to the planes of the slabs. Material diffuses rapidly in along the thin slabs by boundary diffusion and is lost in part by lateral volume diffusion into the thick slabs. Because the boundaries are thin most of the material is lost by lateral diffusion even though $D_b \gg D_l$. The concentration at the original surface is held constant. When volume diffusion is dominant for such an initial condition, the logarithm of the concentration or activity of the diffusing element decreases with the square of the distance from the surface. When boundary diffusion predominates, the logarithm of concentration decreases linearly with distance. Whipple's result is not so simple, but deviates from the linear relation only moderately. In their studies of Ni^{63} diffusion into copper grain boundaries, Yukawa and Sinnott[160] found that D_b from Whipple's equation was 2–5 times that from Fisher's analysis for the same δ. The uncertainty in estimating effective boundary widths and the possibility of boundary motion[161] usually leads to the use of Fisher's simpler equation:

$$\ln\left(\frac{A}{\Delta y}\right) = - \left[\left(\frac{2D_l}{\delta D_b}\right)^{\frac{1}{2}} / (\pi D_l t)^{\frac{1}{4}}\right] y + \text{constant} \quad . \quad (39)$$

where A is the amount of material in a section of thickness Δy parallel to, and displaced by y from, the initial surface and t is the duration of the diffusion anneal.

One of the possible discrepancies between the alternate-slab model and a real polycrystalline specimen has been shown by Barnes.[161] During interdiffusion of copper and nickel he observed strong preferential diffusion along the nickel grain boundaries. At the same time it was evident that the grain boundaries themselves migrated, leaving their copper-rich cusps behind to some extent.

The Burgers model for a tilt boundary implies a high anisotropy of diffusion at low angles of disorientation between neighbouring grains. Evidence confirming this anisotropy led Smoluchowski[162] to propose that three distinguishable kinds of grain-boundary structure are to be expected, depending on the disorientation. Burgers's model should apply from 0° to about 15°. From 15° to about 35°, rod-like distorted regions should result from the bunching of dislocations. The anisotropy should diminish but not vanish. From 35° to 45° the rods should coalesce into large islands of misfit separated by regions of good fit. The anisotropy should be gone. Hoffman[163] found that the anisotropy of grain-boundary self-diffusion in silver was very high at low angles, dropping continuously as the angle increased, but not vanishing at 45°. Thus, the grain boundary must retain a directional structure at maximum disorientation. It may be significant, of course, that the adjoining grains had parallel <100> directions. Okkerse, Tiedema, and Burgers[164] found that grain-boundary self-diffusion in lead varied with disorientation and also with direction in the boundary itself.

There are scattered pieces of evidence relating to rapid diffusion along isolated imperfections or grain boundaries in covalent crystals (e.g., see ref. 137), but very little information is available for such phenomena in ionic crystals. Some of the work on spinels cited earlier was carried out on sintered compacts. In spite of very low volume diffusivities there is little indication that grain-boundary contributions were comparable in total transport. Grain-boundary distributions appear to be as important to the properties and reactions of ceramics as they are to the properties and reactions of metals. Study of grain-boundary diffusivities may be slow in coming only because of the difficulties in preparing pure, sound materials of different grain structures.

VII.—Conclusion

Because of space limitations, various topics which are at present under active study have been omitted from this account. The author wishes to call attention to several of these items* by means of this concluding list:

(1) Accelerated diffusion near the melting point (see the work of Drickamer *et al.* and Nachtrieb *et al.*).

(2) The diffusion of dilute substitutional alloying elements and the effects of dilute impurities on self-diffusion (Lazarus *et al.*, Hoffman and Turnbull, Swalin, Fumi, Overhauser, &c.).

(3) The separation of heats of formation and motion of vacancies by means of quenching-in and annealing-out experiments (Koehler *et al.*, Bradshaw and Pearson, Meechan *et al.*).

(4) The study of diffusion rates by internal-friction measurement (Wert *et al.*, Nowick, LeClaire).

(5) The loci and kinetics of formation of Kirkendall porosity; changes in dimensions of diffusing regions. (Balluffi and Alexander, Barnes, Seith, Heuman, da Silva and Mehl) (see ref. 4 for an extended discussion of this topic).

(6) Diffusion in anisotropic crystals (Shirn, Huntington) and the information they give about mechanisms (dual mechanisms).

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* Several of these topics are discussed by Shewmon.[14]

Appendix

Self-Diffusion in Solid Elements

Element	Type of Diffusion	D_0, cm.²/sec.	Q, k.cal./g.-atom	Ref.
Li		—	9·8	165
		—	9·3	166
		0·23	13·2	167
Na		0·24	10·45	78
		—	9·5	168
		—	9·1	166
		0·36	10·40	167
K		—	9·1	166
Cu		0·6	49	169
		0·08	46·8	170
		0·2	47·12	171
Ag		0·89	45·95	172
		0·40	44·09	173
		0·27	43·70	56*a*
	Grain-boundary	0·03	20·2	154
	Surface	0·16	10·3	146
Au		0·091	41·7	174
		0·14	42·9	175
Fe (α)		5·8	59·7	33
		$2{\cdot}3 \times 10^3$	73·2	32
		530	67·1	50*a*
(γ)		5·8	74·2	32
		0·58	67·9	33
		0·7	68·0	176
Co		0·83	67·7	177
Ni		1·27	66·8	178
Pt		0·33	68·2	190
Pb		0·281	24·21	179
		—	25·7	180
	Grain-boundary	0·81	15·7	180
W		$6{\cdot}3 \times 10^7$	135·8	181
Ta		2	110	191
		$1{\cdot}3 \times 10^3$	110	192
Mg	$\parallel$	1·0	32·2	182
	$\perp$	1·5	32·5	182

APPENDIX—*continued*

Self-Diffusion in Solid Elements

Element	Type of Diffusion	D_0, cm.^{2}/sec.	Q. k.cal./g.-atom	Ref.
Zn	$\parallel$ $\perp$ $\parallel$ $\perp$ Grain-boundary	0·08 0·39 0·13 0·58 0·3	22·0 24·9 21·8 24·3 14·5	183 183 184 184 185
Cd	$\parallel$ $\perp$ Grain-boundary	0·05 0·10 1·0	18·2 19·1 13·0	186 186 186
In (tetragonal)	Average	1·02	17·9	187
Tl (h.c.p., 150°–275° C.) (b.c.c. > 230° C.)	$\parallel$ $\perp$	0·4 0·4 0·7	22·9 22·9 20·0	188 188 188
Sn	$\parallel$ $\perp$	$1·2 \times 10^{-5}$ $3·2 \times 10^{-8}$	10·5 5·9	71 71
Bi	$\parallel$ $\perp$	10^{-3} 10^{46}	31 140	189 189
Ge		7·8	68·5	123
P		$1·07 \times 10^{-3}$	9·4	118
(>30° C.)[$1·07 \times 10^{-3} \exp(-9400/RT) + 2 \times 10^{46} \exp(-80{,}600/RT)$]				
S	$\parallel$ $\perp$	$8·3 \times 10^{-12}$ $1·78 \times 10^{36}$	3·08 78	117 117

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